

CLARK ABOURJEILY

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EDUCATION

Georgia Institute of Technology <i>Atlanta, GA</i>	08/2024 – 05/2026
- M.S. in Robotics, with Core Areas in Controls, Artificial Intelligence (AI), and Perception, GPA 4.0/4.0 - Coursework: Robotics, Nonlinear & Networked Control, Deep Reinforcement Learning, Computer Vision	
American University of Beirut (AUB) <i>Beirut, Lebanon</i>	08/2020 – 06/2024
- B.Eng. in Electrical and Computer Engineering with a Minor in Business Administration, GPA 4.0/4.0 - Graduated with High Distinction, with a Track in Control, Robotics, and Intelligence Systems, ranked 1st - Coursework: Autonomous Mobile Robotics, Machine Learning, Modern Control Systems, Instrumentation	

EXPERIENCE

FACTS Lab, Georgia Tech, Research Assistant <i>Atlanta, GA</i>	05/2025 – Present
- Integrating end-to-end 6-DOF DENSO industrial manipulators for ROS2-based multi-agent collaboration - Enabling safety-critical bimanual planning and control with Signal Temporal Logic (STL) constraints - Developing a VLM-guided (GPT-4o) repair pipeline for infeasible MAPF tasks with MILP (Gurobi) - Engineering RRT-based trajectory execution for robot fleets with evolutionary LLM prompt tuning	
STAR Lab, Georgia Tech, Research Assistant <i>Atlanta, GA</i>	01/2025 – 04/2025
- Developed a motion planner using nonlinear optimization for heterogeneous multi-robot coordination - Added a coalition-aware abstraction to improve synchronization control and eliminate inter-robot collisions	
American University of Beirut (AUB), Teaching Assistant <i>Beirut, Lebanon</i>	09/2022 – 05/2024
- Cooperated with professors in teaching Electronics I & II recitations to junior ECE students - Enhanced concept delivery and graded weekly assignments and quizzes for 50+ students	
Vision and Robotics Lab (VRL), AUB, Research Intern <i>Beirut, Lebanon</i>	06/2023 – 08/2023
- Generated a novel multi-sensor, asynchronous, real, and collaborative 3D reconstruction dataset - Conducted motion capture, experiment design, and sensor fusion using C#, OptiTrack Motive, Cyclone 3DR - Co-authored a research paper for IROS, 2nd author, Didymos-XR Project	

PROJECTS

Look-ahead Unified Camera-LiDAR Inference and Diffusion (LUCID) <i>Atlanta, GA</i>	09/2025 – Present
- Built a multimodal DiT pipeline for dynamics-conditioned future 3D LiDAR generation on nuScenes data - Learned horizon-varying scene forecasts for dynamic autonomous driving environments	
RL-Guided Decentralized Heterogeneous Ant Colony Optimization (ACO) <i>Atlanta, GA</i>	02/2025 – 04/2025
- Built a decentralized ACO framework for multi-agent cooperation on Georgia Tech's Robotarium - Integrated reinforcement learning (SAC) to balance exploration and connectivity for faster convergence - Enforced network connectivity via control barrier functions (CBF) implemented as quadratic programs (QP)	
Implementation and Adaptation of Diversity Is All You Need (DIAYN) in MuJoCo <i>Atlanta, GA</i>	02/2025 – 04/2025
- Reproduced and validated the base DIAYN algorithm across multiple benchmarks (e.g. hopper, cheetah) - Extended DIAYN to a 6-DOF Panda Arm, augmented with HRL/VAE to learn more human-relevant skills	
The Robot Collective: Traffic-Aware Multi-Agent A* Path Planning, Georgia Tech VIP <i>Atlanta, GA</i>	08/2024 – 12/2024
- Improved simulator efficiency of graph-based planner with dynamic edge weighing for block-stacking bots - Mentored undergraduate team members to familiarize them with ROS and to enhance workflow	
Mixed Reality Virtual Testbed for HoloLens-Robot Multi-Agent System <i>Beirut, Lebanon</i>	09/2023 – 05/2024
- Developed a Unity3D AR application for human-robot collaborative operation through ROS 2 and MoveIt - Augmented the physical mobile robot with a virtual robotic manipulator to cut simulation costs	
SLAM, Kinematics and Motion Planning for Ground and Aerial Vehicles using ROS2 <i>Beirut, Lebanon</i>	10/2023 – 12/2023
- Devised an occupancy grid mapping and path planning algorithm for differential drive robot navigation - Developed the autonomous motion control model of a 6-DOF drone	

EXTRACURRICULAR ACTIVITIES

L'Art de la Musique (School of Music) <i>Beirut, Lebanon (Currently Remote, Atlanta, GA)</i>	01/2007 – Present
- Level VIII of Piano Performance, certified by The Lebanese National Higher Conservatory of Music - Gave Classical Piano, Music Theory, and Aural Skills courses to students of ages 6 to 30+	
AUB Robotics Club (AUBRC), Member <i>Beirut, Lebanon</i>	10/2020 – 05/2024
Assisted and participated in several workshops for ROS, Arduino, and Raspberry Pi	
Build It Weekend 5.0 AUBRC x IEEE AUB Student Branch, Robotics Competition <i>Beirut, Lebanon</i>	03/2023 – 04/2023
Designed and built an ESP32-based mobile robot for rescue-task scenarios with IoT control (2nd place)	

SKILLS AND INTERESTS

Languages	English (Fluent), French (Fluent), Arabic (Fluent), Korean (Basic)
Programming	Python, C++, C#
Robotics & ML	ROS, ROS 2, PyTorch, OpenCV, MoveIt2, Gazebo, RViz, MuJoCo
Tools	Linux, Git, GitHub, LaTeX, MATLAB, LabVIEW
Interests	Ski, Tennis, Classical Piano, Robotics, Travel